Module 04

Introduction to DBMS/1

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Module Recap

- Comparison of data management using Python & files and DBMS
- Efficacy and Efficient DBMS highlighted

Module Objectives

- To familiarize with the basic notions and terminology of database management systems
- To understand the role of data models and languages
- To understand the approaches to database design

Module Outline

- Levels of Abstraction

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- Schema & Instance

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- Data Models

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- Relational Databases
- DDL & DML
- SQL
- Database Design

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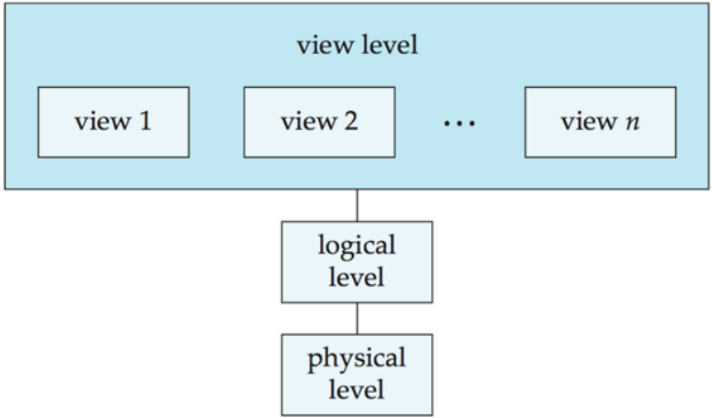
View Level

- **View level:** application programs hide details of data types

View of Data (An architecture for a database system)



Database Management Systems

Partha Pratim Das

Schema and Instance

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Customer Instance

Name	Customer ID	Account #	Aadhaar ID	Mobile #
Pavan Laha	6728	917322	182719289372	9830100291
Lata Kala	8912	827183	918291204829	7189203928
Nand Prabhu	6617	372912	127837291021	8892021892

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Account Instance

Account #	Account Type	Interest Rate	Min. Bal.	Balance
917322	Savings	4.0%	5000	7812
372912	Current	0.0%		291820
827183	Term Deposit		10000	100000

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Design Principle

In general, the interfaces between the various levels and components should be well defined so that changes in some parts do not seriously influence others.

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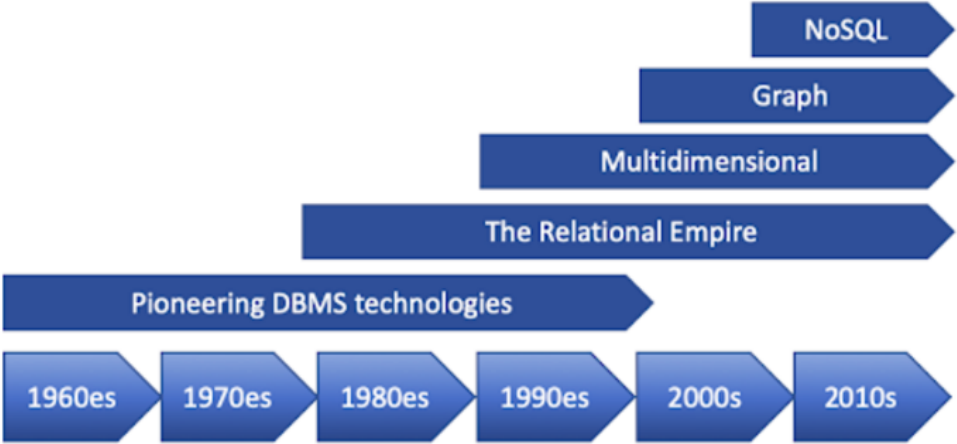
  - Content Addressable Storage (CAS) with metadata descriptors
  - XML format
  - DBMS which supports BLOBs

Data Models (2)



Relational Model

- All the data is stored in various tables

(a) The instructor table

ID	name	dept_name	salary
22222	Einstein	Physics	95000
12121	Wu	Finance	90000
32343	El Said	History	60000
45565	Katz	Comp. Sci.	75000
98345	Kim	Elec. Eng.	80000
76766	Crick	Biology	72000
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- Example of tabular data in the relational model

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A Sample Relational Database

(a) The instructor table

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...	...	...	...

(b) The department table

dept_name	building	budget
Comp. Sci.	Taylor	100000
Biology	Watson	90000
Elec. Eng.	Taylor	85000
Music	Packard	80000
Finance	Painter	120000
History	Painter	50000
Physics	Watson	70000

DDL and DML

Data Definition Language (DDL)

- Specification notation for defining the database schema

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create table instructor (
    ID          char(5),
    name        varchar(20),
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  - Authorization (Who can access what)

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  - Language extensions to allow embedded SQL
  - Application Programming Interface or API (for example, ODBC/JDBC) which allow SQL queries to be sent to a database

Database Design

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  - **Business decision:** What attributes should we record in the database?
  - **Computer Science decision:** What relation schemas should we have and how should the attributes be distributed among the various relation schemas?
- **Physical Design** – Deciding on the physical layout of the database

Database Design (2)

Is there any problem with this relation?

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Module Summary

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